barix

Barix AG Firmware Attack Surface & Detection Brief

Scope & status (2026-10-27). This vendor was on the "46 genuine OT vendors still missing firmware" list and was acquired via **VirusTotal Intelligence** during the 2026-10-26 acquire-and-analyze pass. The **behavioural differential is complete** (see the section below). Full architecture enumeration, service mapping, and CVE/advisory research have **not yet been performed** — unlike the earlier CVE-researched briefs, this brief currently carries the VT behavioural differential plus product/Purdue scoping only. Anything not stated below is "not yet researched," not "absent."

Census exposure: 4 OT & Building-Systems devices across 3 states (per vendor-exposure-vs-coverage.md). A small footprint, acquired for completeness.

What Barix makes. Barix AG is a Swiss manufacturer of **audio-over-IP** endpoints: the *Exstreamer* (IP audio decoder/player), *Instreamer* (audio encoder), *Annuncicom* (IP intercom / paging device with GPIO relay outputs and contact inputs), and the IPAM audio module family. In practice these appear in **IP public-address, mass-notification ("giant voice"), intercom, and background-audio** systems. The higher-end devices run **embedded Linux** on ARM/MIPS — which is why this vendor's baseline has a meaningful **ELF native-import plane** — while discovery and configuration tooling (the Barix Discovery Tool / firmware uploader) is Windows-side.

Purdue layer mapping. The audio endpoints are **field devices (L1-equivalent by role)** — a direct physical-world interface (audio output, GPIO relay/contact I/O) with no intervening controller, the same role framing used for Axis cameras and the A1001 door controller. The Windows-side discovery/config tooling is management-plane (**L2/L3**). An Annuncicom driving relay outputs on a notification circuit is the audio analog of a door-strike relay: compromise is a **physical-effect** concern (spoofed or suppressed announcements), not only a data concern.

Data-quality note. The VT-Intelligence acquisition for Barix is heterogeneous — embedded-Linux firmware images alongside ancillary files. The single-engine `trojan.`-labelled ELF samples that appear in `output/new-vendor-vt-detections.csv` for this vendor are consistent with the project-wide finding that **single-engine detections on embedded firmware are false-positive noise**, not confirmed malware; none reached the >=5-engine bar used for the impersonation split. Treat the baseline as sound but noise-bearing until a dedicated acquisition review is done.

VT behavioural differential — Barix AG corpus (2026-10-27)

Generated by `tools/Invoke-VendorDifferential.ps1` against the Barix AG known-good baseline (NSRL + local, platform-matched) and the VT malicious target set (7913 targets). Each behaviour is scored by how **rare** it is in this vendor's own known-good software: `rarity 100` = the baseline *never* exhibits it; a lower score = the baseline does, so it is scored down as benign-common.

> **Read this correctly.** The sparse OT baseline means the rarity-100 list is dominated by *generic* malicious behaviour, not Barix AG-specific indicators — any comparable malware trips most of it. The genuinely Barix AG-specific product of this differential is the **suppression list** at the end: behaviours this vendor's real software performs, which must **not** be alerted on when monitoring a host that runs it. Use the MITRE table as the TTP-tier hunting frame; use the suppression list to cut false positives.

Category	Behaviours observed	Discriminating (rarity 100)	Benign-common (suppressed)
MITRE ATT&CK	852	852	0
Windows API	210	131	79
Process/cmdline	6164	6164	0
Registry	43285	43285	0
Mutex	1898	1898	0
YARA	1460	1458	2
Sigma	588	585	3
Certificate	4	3	1
Tags	263	211	52
ELF (native imports)	1186	655	531

Top discriminating MITRE ATT&CK techniques

Techniques present in the malicious corpus and **absent from the Barix AG known-good baseline**, ranked by how many malicious samples exhibit each.

Technique	Malicious samples	Last seen
T1082: Reads software policies	404	2026-02-04
T1082: Queries the volume information (name, serial number etc) of a device	350	2026-02-04
T1071: Adversaries may communicate using application layer protocols to avoid detection/network filtering by blending in with existing traffic.	348	2026-02-04
T1083: Reads ini files	321	2026-02-04
T1497: May sleep (evasive loops) to hinder dynamic analysis	319	2026-02-04
T1036: Creates files inside the user directory	289	2026-02-04
T1010: Sample monitors Window changes (e.g. starting applications), analyze the sample with the simulation cookbook	266	2026-02-04
T1518.001: May try to detect the virtual machine to hinder analysis (VM artifact strings found in memory)	266	2026-02-04

Technique	Malicious samples	Last seen
T1574.002: Tries to load missing DLLs	266	2026-02-04
T1497: Contains long sleeps (>= 3 min)	263	2026-02-04
T1082: An adversary may attempt to get detailed information about the operating system and hardware, including version, patches, hotfixes, service packs, and architecture.	247	2026-02-04
T1497: Contains medium sleeps (>= 30s)	238	2026-02-04

Highest-prevalence behavioural discriminators (top 5 per type)

Behaviour	Type	Malicious samples
HKEY_LOCAL_MACHINE\Software\Microsoft\OLE	Registry	421
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\SessionIn	Registry	316
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\	Registry	314
HKEY_CURRENT_USER\software	Registry	307
HKEY_LOCAL_MACHINE\Software\Microsoft\Fusion	Registry	290
C:\Windows\system32\services.exe	Process/cmdline	299
C:\Windows\System32\conhost.exe C:\Windows\system32\conhost.exe 0xffffffff -Fo	Process/cmdline	287
C:\Windows\Explorer.EXE	Process/cmdline	216
C:\Windows\System32\svchost.exe -k NetworkService -p	Process/cmdline	130
C:\Windows\system32\svchost.exe -k UnistackSvcGroup	Process/cmdline	130
kernel32.dll!GetLocalTime	Windows API	9
kernel32.dll!GetFileTime	Windows API	8
kernel32.dll!FindAtomW	Windows API	7
kernel32.dll!CopyFileA	Windows API	6
kernel32.dll!CopyFileExA	Windows API	6
Local\ZonesLockedCacheCounterMutex	Mutex	229
Local\ZonesCacheCounterMutex	Mutex	227
Global\MTX_MSO_AdHoc1_S-1-5-21-1547548985-192504236-2982455695-500	Mutex	146
Global\MTX_MSO_Formal1_S-1-5-21-1547548985-192504236-2982455695-500	Mutex	146
\Sessions\1\BaseNamedObjects\Local\ZonesCacheCounterMutex	Mutex	132

Device-firmware plane — top ELF native-import discriminators

The Barix AG baseline also covers embedded-Linux device firmware; these ELF imports appear in the malicious ELF corpus but not in Barix AG firmware.

Symbol / import	Malicious samples
ELF!_Jv_RegisterClasses	260
ELF!__deregister_frame_info	135
ELF!__register_frame_info	132
ELF!__pthread_initialize_minimal	112
ELF!htons	59
ELF!_gp_disp	32
ELF!__nptl_deallocate_tsd	29
ELF!__nptl_nthreads	29

Benign-common behaviours — suppress for this vendor (do NOT alert)

These appear in the Barix AG known-good baseline, so the differential scores them down and they should not drive alerts on a host running genuine Barix AG software. `base%` = share of the baseline exhibiting the behaviour. (663 suppressed in total; top 15 shown.)

Behaviour	Type	base%	rarity
No Sig	Certificate	99.9832%	0
pyc	Tags	39.8518%	60
text	Tags	14.7044%	85
elf	Tags	12.8179%	87
known-distributor	Tags	10.2914%	90
arm	Tags	9.0281%	91
ELF!**gmon_start**	ELF (native imports)	7.3438%	93
python	Tags	6.6869%	93
ELF!_ITM_deregisterTMCloneTable	ELF (native imports)	6.03%	94
shared-lib	Tags	5.9458%	94
ELF!memcpy	ELF (native imports)	5.0531%	95
ELF!_ITM_registerTMCloneTable	ELF (native imports)	4.9183%	95
detect-debug-environment	Tags	4.6657%	95
ELF!memset	ELF (native imports)	4.3456%	96
ELF!free	ELF (native imports)	4.0088%	96